Olivia Santiago

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Active Security Clearance Level: Interim Secret

SUMMARY

Fourth-year Mechanical Engineering student at Cornell University seeking full-time roles for 2027, with experience designing, analyzing, manufacturing, and testing aerospace flight hardware through AFRL internships and Cornell's UAV and satellite teams. Experienced in developing mechanical systems from requirements through fabrication, integration, and validation, with interests in structural design, deployable mechanisms, manufacturability, and hardware testing.

EDUCATION

Cornell University, Duffield College of Engineering **I*thaca*, NY**
Bachelor of Science in Mechanical Engineering *Expected May 2027*
GPA: 3.67/4.30
Relevant Coursework: Mechanics of Materials, System Dynamics, Fluid Mechanics, Dynamics, Spaceflight Mechanics, Statics & Mechanics of Solids, Thermodynamics, Physics (I-III), Linear Algebra, Differential Equations, Multivariable Calculus

Universidad de Cantabria **Santander, Spain**
Engineering Exchange Program *January 2026 – June 2026*
Relevant Coursework: Heat Transfer, Uncertainty Analysis in Engineering, Industrial Robotics & Computer Vision

WORK EXPERIENCE

Air Force Research Laboratory Scholars Program at Kirtland AFB ***Albuquerque, NM***
AFRL Scholar – Mechanical Engineering Intern *June 2026 – August 2026*
- Designed and manufactured a gravity offload follower prototype to support active testing of deployable satellite technology in the Deployable Structures Lab (DeSel)
- Modeled and managed a 5-subassembly SolidWorks assembly, iterating designs based on structural analysis and drawings based on manufacturability feedback from machinists
- Performed static structural analysis in ANSYS to validate that the structure supports 50 lb. deployable offload points
- Collaborated with a 3-person intern team and a research mentor to translate design requirements into a functional prototype
- Presented the design in 3 formal design reviews to AFRL engineers and government contractors
- Led a demonstration of the prototype for the government lead and prepared a demonstration for AFRL commander tour

Air Force Research Laboratory Scholars Program at Edwards AFB ***Edwards Air Force Base, CA***
AFRL Scholar – Mechanical Engineering Intern *May 2025 – August 2025*
- Took ownership of a full-scope research project, independently leading reverse-engineering of adversary UAV performance using open-source intelligence, enabling AFRL to benchmark foreign system capabilities against U.S. platforms
- Designed MATLAB-based dynamic trajectory simulation tools to model 3DOF flight dynamics and performance envelopes
- Integrated aerodynamic and propulsion data into simulations to evaluate climb performance and turn rate, producing results informing AFRL's threat assessment models
- Presented research findings to 20+ AFRL engineers, researchers, and industry partners

PROJECT EXPERIENCE

Cornell Unmanned Aerial Systems Project Team (CUAir) at Cornell University ***Ithaca, NY***
Integration & Testing Sub team Lead *September 2024 – Present*
- Direct a 6-person sub team designing, manufacturing, and testing critical UAV subsystems, including the electrical bay, control surfaces, composite materials testing, wing loading, and tilt-rotor test stand, overseeing deliverables from concept through flight-ready implementation
- Engineered the UAV nose section's electrical bay for integration of dual 6s batteries on a 55 lb. platform, reducing battery box weight by 3%, and accommodating complex geometry by designing custom 3D-printed components and laser-cut wood panels in SolidWorks, collaborating closely with the electrical team; survived all test flights
- Converted an 1800 mm wingspan fixed-wing Skyhunter kit plane into a mechanically reinforced VTOL testing platform for a 35 lb. UAV, designing and installing front carbon fiber booms, VTOL motor mounts, a side access panel, and modular wing joiners, along with custom add-ons to organize and improve accessibility of electrical components

Space Systems Design Studio at Cornell University ***Ithaca, NY***
Alpha CubeSat Integration & Testing *January 2025 – June 2025*
- Conducted thermal testing of the Alpha CubeSat in a vacuum chamber while monitoring component temperatures

- Performed light sail deployment and structural tests to validate flight hardware for mission success
- Supported cleanroom assembly and final integration of the Alpha CubeSat hardware for launch aboard SpaceX NG-23

DeSCENT ChipSat Deployer Machinist *September 2024 – December 2024*

- Machined precise aluminum components for the DeSCENT ChipSat Deployer with end mills up to 1/8" in diameter to ±.005" in tolerance, preparing the system to deploy 100 ChipSats on a Blue Origin suborbital mission

SKILLS

Design: SolidWorks, Fusion 360, MATLAB, Python, ANSYS Mechanical, R, RobotStudio
Manufacturing: GD&T, Rapid Prototyping, 3D Printing, Laser Cutting, Machining (Mill & Lathe), Composite Layups

INTERESTS/ACTIVITIES

French Horn – Cornell Symphony Orchestra
Golfer – Competitive golfer for 13 years; playing since age 5
Solo Backpacking – Traveled independently through 7 countries and 12 cities